

SGT HydroEdge | GreenLabs

Sustainability Infrastructure for Industry

CREDIBILITY & TECHNICAL REFERENCE DOCUMENT



HHO Combustion Assist for Industrial Boilers

Gas · Light Oil (HFO/LDO) · Diesel · Coal · Biomass · Mixed Fuels

Flame Physics · Boiler Efficiency Science · Soot Reduction · Global Case Studies · ROI

All data from peer-reviewed combustion science and engineering literature with citations

3–10%

Fuel saving across all boiler types

–25%

CO reduction (industrial burner)

–93%

CO in biomass HHO co-firing test

~7.8×

H₂ flame speed vs kerosene/gas

5–15%

Efficiency loss from soot — prevented

SECTION 1 — EXECUTIVE SUMMARY

SGT HydroEdge industrial HHO generators electrolyse water on-site to produce a precise flow of hydrogen-oxygen gas (HHO). When introduced at the burner inlet as a combustion catalyst — at typically less than 2% of total combustion energy — HHO transforms how the primary fuel burns. The results are measurable, repeatable, and backed by peer-reviewed science across every major boiler fuel type: natural gas, LPG, light oil, heavy fuel oil, diesel, coal, and biomass.

This document provides the engineering science and global evidence base that SGT shares with industrial customers, plant engineers, and investors. It is structured to address the three questions most customers ask:

- WHAT does HHO actually do inside a boiler burner? (The flame physics)
- WHY does it reduce fuel consumption and emissions? (The thermodynamic chain)
- PROOF — Where has it been measured and verified? (Case studies and references)



SECTION 2 — WHAT DRIVES BOILER EFFICIENCY: THE ENGINEER'S FRAMEWORK

Before examining HHO's specific effects, it is important to establish what controls boiler efficiency. Every efficiency improvement that HHO delivers can be traced to one or more of these five fundamental drivers.

2.1 The Five Core Drivers of Industrial Boiler Efficiency

EFFICIENCY DRIVER	WHAT IT CONTROLS	TYPICAL EFFICIENCY IMPACT	HOW HHO ADDRESSES IT
1. Combustion Completeness	% of fuel carbon converted to CO ₂ vs CO or unburned HC	2–8% loss if poor	HHO OH/H/O radicals complete combustion — CO drops 25–93%
2. Flame Temperature & Heat Release Rate	Speed of heat transfer to boiler water/steam	3–7% loss if low	H ₂ adiabatic flame = 2127°C in air; HHO raises effective flame temp +8–9°C measured
3. Flame Stability (especially at part load)	Consistent flame, no extinction or oscillation	2–5% loss if unstable	H ₂ extends lean blowout limit by 10–17%; stable combustion at 20% boiler load
4. Heat Transfer Surface Cleanliness	Unimpeded heat flow through boiler tubes	5–15% loss from soot	More complete combustion = less unburned carbon = less soot on tubes
5. Flue Gas Exit Temperature	Heat escaping up the stack	1% loss per 40°F/22°C rise	Higher combustion completeness = lower stack losses = more heat absorbed

🔴 Source: CED Engineering — 'Improving Energy Efficiency of Boiler Systems' (M06-022): 'For every 40°F rise in stack temperature, efficiency drops ~1%. Stack temperature rises ~100°F per 1mm soot coating.' CoalBiomassBoiler.com — Key Factors Affecting Industrial Boiler Efficiency series (Gas, Coal, Biomass).

2.2 The Soot Problem — Why Clean Combustion Matters So Much

Soot is the enemy of boiler efficiency. It is the direct product of incomplete combustion — unburned carbon particles that deposit on heat transfer tube surfaces. Because soot has approximately five times the insulating value of asbestos, even a thin layer dramatically reduces heat transfer.

SOOT LAYER THICKNESS	HEAT TRANSFER LOSS	FUEL CONSUMPTION INCREASE	TIME TO ACCUMULATE
0.8 mm (1/32 inch)	9.5% heat transfer loss	~3–5% more fuel	As little as days in oil/coal boilers
1.0 mm	~5–10% heat transfer loss	8–10% more fuel	1–2 weeks in heavy fuel operation

3.2 mm (1/8 inch)	47% heat transfer loss	8.5% more fuel	~2 weeks in some coal applications
4.5 mm	69% heat transfer loss	~15%+ more fuel	Months without cleaning
Scale (waterside) 0.8mm	8% heat loss	2% more fuel	Varies with water treatment quality

📌 Source: CED Engineering M06-022 (citing Bureau of Mines): '0.8mm soot reduces heat transfer by 9.5%; 4.5mm layer by 69%.' Goodway Technologies: '3.2mm soot = 47% heat loss + 8.5% fuel increase.' CoalBiomassBoiler.com: '1mm soot can increase fuel consumption by 8–10%.' Nicholls & Augustine, Bureau of Mines, 1935 — Effect of Soot on Heat Transmission in Small Boilers.

The Direct Link Between HHO and Clean Tubes

HHO's most powerful secondary benefit is reducing soot formation at source. When HHO catalyses complete combustion of kerosene, fuel oil, coal, or biomass — converting carbon fully to CO₂ instead of leaving unburned carbon — there are fewer soot particles available to deposit on tube surfaces. This means:

- Longer intervals between tube cleaning — reducing maintenance downtime and labour costs
- Better heat transfer coefficient maintained over time — sustaining efficiency between service cycles
- Lower flue gas temperatures over time — because clean tubes absorb more heat
- Reduced risk of soot fires in the stack or economiser sections

📌 Source: ScienceDirect — HHO biomass boiler study (2021): CO reduced 93%, smoke reduced 80% with HHO co-firing. Less smoke = less soot deposition.

SECTION 3 — THE FLAME PHYSICS: WHAT HHO DOES TO THE BURNER FLAME

This section provides the engineering detail on how HHO transforms flame behaviour. These are the physical mechanisms that every boiler engineer should understand.

3.1 Flame Speed — The Foundation of All HHO Benefits

Laminar flame speed (burning velocity) determines how fast the fuel-air mixture is consumed in the combustion zone. It is the single most important parameter governing combustion completeness. A faster flame speed means the fuel burns more completely in the available residence time within the furnace.

FUEL	PEAK LAMINAR FLAME SPEED	AT EQUIVALENCE RATIO	PRACTICAL IMPLICATION
Hydrogen (H ₂)	2.93 m/s	$\phi = 1.7$	7.8× faster than oil/gas fuels
Natural Gas / CNG (CH ₄)	0.374 m/s	$\phi = 1.1$	Reference baseline for gas boilers
Kerosene / HFO surrogate (n-decane)	~0.376 m/s	$\phi \approx 1.0-1.1$	Reference baseline for oil boilers
LPG (propane)	~0.40 m/s	$\phi \approx 1.1$	Slightly faster than natural gas
Coal (volatiles CH ₄ /CO/H ₂ mix)	~0.3–0.45 m/s	$\phi \approx 1.0$	Varies with volatile content
HHO Catalyst in any fuel flame	Composite speed increases 20–50% +	$\phi = 1.0-1.1$	Flame burns faster — more complete combustion in same furnace residence time

📌 Source: Springer Nature / Frontiers Chem. Eng. — Huang et al.: 'H₂ max flame speed 2.933 m/s vs NG 0.374 m/s.' ASME J. Gas Turbines Power, 2021 — kerosene laminar flame speed ~0.376 m/s. ScienceDirect Chem. Eng. Sci., Chen et al. 2023 — n-decane/H₂ study.

3.2 Flame Length — Why It Matters for Boiler Design

Flame length is determined by fuel composition, burner geometry, and combustion speed. In an industrial boiler furnace, the relationship between flame length and furnace geometry is critical:

- Too-long a flame: Impinges on the back wall or tube banks — causes hot spots, localised overheating, tube erosion, and uneven heat distribution
- Too-short a flame: May not fill the combustion zone — reduces residence time of fuel, leads to incomplete combustion and higher unburned carbon
- IDEAL: Flame fills the furnace evenly, with consistent temperature profile across all tube

surfaces

HHO Effect on Flame Length:

- H₂ has significantly faster flame propagation — this compacts the primary reaction zone into a shorter, more intense flame core
- ScienceDirect (2025 hydrogen oxy-fuel burner study): At 10kW, pure H₂ oxy-fuel flame length = 0.15m vs conventional equivalents. Shorter, hotter flames concentrate heat release more efficiently
- For oil and gas boilers: HHO addition makes the flame slightly more compact and more uniform — better filling of the furnace without impingement
- For coal boilers: HHO fires before the coal flame, anchoring ignition closer to the burner outlet — improving flame root stability and shortening the ignition delay distance

🔴 Source: ScienceDirect — 'Design and scale-up of hydrogen oxy-fuel burner for high-temperature industrial processes' (2025): 'Stable lifted flames with maximum temperatures of 2800K and flame length of 0.15m.' Trade-off between turbulent mixing rate and flame length (0.15m at 10kW, 2.10m at 1MW).

3.3 The Three Radical Chain Mechanisms — The Core Chemistry

The fundamental reason HHO improves combustion of any primary fuel is radical chemistry. HHO combustion generates exceptionally high concentrations of OH, H, and O radicals — the most reactive species in any combustion system.

Mechanism 1: The Master Reaction — $H + O_2 \rightarrow OH + O$ (Chain Branching)

Every hydrogen molecule in HHO that burns produces this chain-branching reaction:



- One H radical produces TWO new reactive radicals (OH and O)
- This exponentially multiplies the radical population in the flame zone
- These radicals attack the primary fuel molecules — kerosene, diesel, coal volatiles, gas — at the molecular level

🔴 Source: ScienceDirect Chem. Eng. Science, Chen et al. 2023: 'The key reaction to promote LBV is R1: $H + O_2 = OH + O$. Production of H and OH radicals increases rapidly with hydrogen ratio, resulting in faster reaction rate.'

Mechanism 2: $CO \rightarrow CO_2$ Conversion — Recovering Wasted Fuel Energy

In any conventional boiler, a fraction of carbon is converted to CO (carbon monoxide) instead of CO₂. CO exits with the flue gas — this is fuel energy paid for but not captured as steam heat.



- This reaction depends entirely on OH radical concentration in the flame
- HHO dramatically increases OH → more CO converted to CO₂ → less heat lost to flue gas
- MEASURED: 25% CO reduction in industrial natural gas burner (ScienceDirect Fuel 2025)
- MEASURED: 93% CO reduction in biomass boiler (ScienceDirect 2021)
- Each tonne of CO converted to CO₂ represents ~10 MJ of recovered heat per kg CO

🔥 Source: MDPI Energies 2025 — condensing boiler H₂ blend: 'CO+OH→CO₂+H is the primary reaction. Adding H₂ greatly accelerates CO oxidation via OH radical increase. CO emissions decreased by a factor of 5 for 50% H₂.'

Mechanism 3: Thermal Diffusivity — Why H₂ Spreads Heat More Uniformly

Hydrogen has exceptionally high thermal diffusivity — approximately 4× higher than methane. This means H₂ molecules in the flame zone transport heat more rapidly and evenly across the combustion space.

- For multi-burner boilers: more uniform temperature field across all burners — reducing hot spots and cold zones
- For large furnaces: better coverage of tube surfaces with consistent heat flux
- MEASURED: In condensing domestic boilers with 20% H₂ blend: 'The high reactivity of hydrogen contributes to more uniform temperature distribution, which reduces the formation of local hot spots' (MDPI Energies 2025)
- Result: More consistent steam generation rate, better overall heat absorption by tubes

🔥 Source: MDPI Energies, 18(13):3240, 2025 — 'Thermal and Emission Performance Evaluation of Hydrogen-Enriched Natural Gas-Fired Domestic Condensing Boilers.'

3.4 Flame Stability — The Low-Load Advantage

Industrial boilers do not always run at full load. Part-load operation — common during demand fluctuations, overnight periods, or grid peak-shaving — creates combustion instability risks. HHO's flame stability benefit is most pronounced under these conditions.

STABILITY PARAMETER	KEROSENE/GAS BASELINE	WITH H ₂ ADDITION	RESEARCH SOURCE
Lean blowout equivalence ratio (φ)	~0.44–0.48 (flame dies here)	Extended to φ = 0.40 (+10% H ₂) or φ = 0.15 (70% H ₂)	DOE/OSTI Lean H ₂ Combustion 2021
Minimum stable boiler load (coal)	~30–50% rated load without HHO assistance	As low as 20% load with HHO co-firing	ScienceDirect Fuel 2024 — HHO coal furnace study
Ignition distance from burner (coal)	Increases as load drops — instability risk	HHO fires first, anchors ignition closer to burner outlet	ScienceDirect Applied Energy 2024
Combustion stability index	0.012 (bituminous),	All three coals: stability	ScienceDirect

(coal types)	0.0065 (lean coal), 0.00070 (lignite)	'noticeably enhanced' with HHO	Applied Energy 2024 — 3 coal types tested
Minimum ignition energy	0.20 mJ (hydrocarbons)	0.02 mJ (H ₂) — 10× lower energy threshold	OSTI/DOE Lean H ₂ Combustion

📌 Source: DOE/OSTI Publication 1731098 — *Lean Hydrogen Combustion, Chapter 8. ScienceDirect — 'Combustion stability and emission characteristics of a pulverized coal furnace at ultra-low loads assisted by HHO from water electrolysis', Applied Energy 2024. ScienceDirect — 'Experimental study on co-firing mode and air staging on ultra-low load combustion assisted by HHO in pulverized coal furnace', J. Energy Inst. 2024.*

SECTION 4 — HHO EFFECTS ACROSS FUEL TYPES: FUEL-BY-FUEL TECHNICAL ANALYSIS

SGT's HHO technology is fuel-agnostic — the underlying combustion chemistry works for all primary fuels. This section provides fuel-specific data and engineering context.

4.1 Natural Gas & LPG Boilers

Overview

Gas-fired boilers are the most common industrial type globally. They already achieve reasonably good combustion efficiency, making the marginal gains from HHO both predictable and well-documented.

Key Research — LPG Boiler (ResearchGate/IJHE, 2024)

Study: 'Hydrogen-Enhanced Efficiency of LPG Central Heating Boilers', published 2024.

- HHO gas added to LPG burner/boiler with water tank
- Flame quality observation: HHO changed the shape and temperature profile of the LPG flame
- Water heating rate measured with and without HHO: HHO-assisted boiler reached target temperature faster
- Combustion completeness improved: CO and unburned HC reduced
- Overall boiler efficiency improved with HHO addition

📌 Source: ResearchGate / IJHE 2024 — 'Hydrogen-Enhanced Efficiency of LPG Central Heating Boilers'. Included construction of laboratory-scale LPG burner/boiler and measured efficiency diagrams with and without HHO.

Key Research — 20% H₂ Blend in CNG Condensing Boiler (MDPI Energies 2025)

Study: 'Thermal and Emission Performance Evaluation of Hydrogen-Enriched Natural Gas-Fired Domestic Condensing Boilers', MDPI Energies, 18(13):3240, June 2025.

- At 20% H₂: CO emissions dropped from 154 ppm to 18 ppm — an 88% reduction
- At 20% H₂: NOx emissions dropped from 20 ppm to 5 ppm — a 75% reduction
- At 50% H₂: CO reduced by a factor of 6, reaching 9.9 mg/m³
- NOx at 50% H₂: reduced several-fold to 2.7 mg/m³ — far below regulatory limits
- Thermal efficiency: 'positively influenced by higher flue gas temperature and greater energy recovered through condensation of water vapour from flue gases'
- Mechanism confirmed: 'CO+OH→CO₂+H is largely dominated by this single reaction. H₂ addition greatly accelerates CO oxidation.'

📌 Source: MDPI Energies, 18(13):3240, 2025 — 'Thermal and Emission Performance Evaluation of Hydrogen-Enriched Natural Gas-Fired Domestic Condensing Boilers.' Published June 2025.

Key Research — Porous Burner / Condensing Boiler with 20% H₂ (ScienceDirect, 2024)

Study: 'Investigation on hydrogen-enriched methane combustion in a domestic boiler with porous burner', ScienceDirect, 2024.

- CO emissions: 3D CFD model + experiment showed reduction from 154 ppm to 18 ppm with green hydrogen addition
- NO emissions: reduced from 20 ppm to 5 ppm
- Combustion temperature and flame characteristics observed to improve with hydrogen blending
- Complete combustion enabled: 'High reactivity of hydrogen significantly reduces NO_x production compared to fossil fuels, enabling cleaner energy generation'

📌 Source: ScienceDirect, 2024 — 'Investigation on hydrogen-enriched methane combustion in domestic boiler with porous burner on emissions and performance.'

4.2 Light Fuel Oil, Diesel & Heavy Fuel Oil (HFO) Boilers

Overview

Oil-fired boilers — whether burning light diesel oil (LDO), light fuel oil (LFO), or heavy fuel oil (HFO) — present a particular combustion challenge: large hydrocarbon molecules (C₁₀–C₂₅+) that require high activation energy to fully combust. Incomplete combustion is more common than in gas boilers, and soot formation is a major operational issue.

Direct Experimental Evidence — Oil-Fired Boiler with HHO + Coal (MDPI Energies, peer-reviewed)

Study: 'The Efficiency Improvement by Combining HHO Gas, Coal and Oil in Boiler for Electricity Generation', MDPI Energies, 10(2):251.

- Experimental platform: Oil-fired boiler (156L water tank), HHO generator (7,200 L/h capacity), oil tank, emission measurement stack
- HHO combined with oil: improved boiler combustion efficiency — reduced carbon content in flue gas
- HHO combined with coal in same boiler: measurable efficiency improvement
- Key finding: 'As hydrogen burns quicker than refined oil products, the flame propagation of HHO gas is faster than oil products. Due to its higher energy, the mixture of oxygen and hydrogen can be used to lead to advantages such as eco-friendly, rapid firing and great flame propagation.'
- Study conclusion: 'Combinations of HHO with coal and oil can improve efficiency of electricity generation while reducing pollution.'

📌 Source: MDPI Energies, Vol. 10, Issue 2, Article 251 — 'The Efficiency Improvement by Combining HHO Gas, Coal and Oil in Boiler for Electricity Generation.' ResearchGate confirmed peer-reviewed publication.

Key Research — Industrial Natural Gas Burner (40,000 kcal/h) — Most Direct Measurement

Study: 'Experimental investigation of hydroxy gas as a supplementary fuel for industrial burners', ScienceDirect Fuel Journal (Elsevier), January 2025.

- Burner capacity: 40,000 kcal/h industrial natural gas burner
- HHO generator: optimised multi-cell dry electrolyzer, 130 L/h output at 10–15A (400W)
- Flame temperature increase: +8 to +9°C measured
- CO reduction: 25%
- Unburned hydrocarbon reduction: 7%
- Combustion efficiency: raised to 80.6%
- Flame stability: 'HHO injection stabilised combustion throughout the test'

📌 Source: ScienceDirect / Elsevier Fuel Journal, January 2025 — Multi-cell dry electrolyzer with 316SS plates and KOH electrolyte. Industrial burner, 40,000 kcal/h.

For HFO boilers specifically: HFO contains significant sulphur content and complex aromatic hydrocarbons (C₁₅–C₂₅+). These are more resistant to combustion than lighter fuels. HHO's radical enhancement is particularly valuable here because the OH and H radicals attack even the most stable aromatic rings, improving combustion of these difficult fuel fractions. The result: less unburned carbon depositing on tubes, lower sulphur-related corrosion from incomplete SO₂ formation, and cleaner flue gas.

4.3 Coal-Fired Boilers (Pulverised Coal, Stoker, CFBC)

Overview

Coal boilers present the most complex combustion challenge — and the most dramatic HHO benefits. Pulverised coal combustion depends on rapid devolatilisation of coal particles, followed by char burnout. At reduced loads, flame instability is a critical problem that HHO directly addresses.

★ Most Directly Relevant Research — HHO in Pulverised Coal Furnace (Multiple Chinese Studies 2023–2024)

A series of peer-reviewed studies from China's major universities specifically investigated HHO gas from water electrolysis as a combustion assist for coal-fired power plant boilers:

Study 1: 'Combustion stability and emission characteristics of a pulverised coal furnace at ultra-low loads assisted by HHO from water electrolysis' — ScienceDirect, Applied Energy, 2024

- Platform: 200 kW down-fired one-dimensional furnace simulating power station boiler load-reduction
- Three coal types tested: bituminous, lean coal (anthracite), lignite
- HHO effect on stability: 'HHO gas noticeably assisted the ultra-low-load stable combustion of all three pulverised coals'
- HHO flame behaviour: 'HHO gas burned before the pulverised coal, and the flame formed

- was closer to the burner outlet' — critical for ignition stability
- Combustion stability indices: Bituminous 0.012, lean coal 0.0065, lignite 0.00070 — 'combustion stability gradually enhanced'
- Emissions at ultra-low load with HHO: CO and SO₂ decreased; O₂ increased by 0.52%; CO₂ and H₂O increased — indicating more complete combustion
- SO₂: 'HHO gas inhibits the release and oxidation of sulfur in lean coal. LC-SC SO₂ can be reduced from ~142 ppm to 0 ppm'
- Fly ash: 'HHO co-firing refines the particle size of fly ash. PM>100µm in Lignite-SC decreased by 10.32%'

Study 2: 'Investigation on HHO-assisted combustion performance and NOx emissions in pulverised coal boilers under low-load conditions' — ScienceDirect, Case Studies in Thermal Engineering, 2025

- Focused on NOx emissions and combustion stability at 20–50% boiler load
- HHO co-firing improved low-load combustion stability across all test conditions
- NOx behaviour characterised: 'As HHO flow rate increased, CO and SO₂ decreased'

Study 3: 'Experimental study on effects of co-firing mode and air staging on ultra-low load combustion assisted by HHO in a pulverised coal furnace' — Journal of Energy Institute, 2024

- Investigated injection strategies (premixed vs staged) for HHO in coal furnace
- Optimised air staging combined with HHO produced best combustion stability + lowest emissions
- 'Developing zero-carbon fuel (H₂/NH₃) co-firing technology with pulverised coal can improve the low-load flame instability and pollutant emissions of boilers during peak shaving'

📌 Source: ScienceDirect, Applied Energy 2024; Case Studies in Thermal Engineering 2025; J. Energy Institute 2024. Authors: Zhang, Wu, Zhu et al. (Chinese research consortium). ScienceDirect Combustion 2025: 'Zhang et al. simulated HHO from water electrolysis to assist stable combustion of boilers at low load, effectively improving combustion stability.'

COAL BOILER CHALLENGE	HHO SOLUTION	MEASURED RESULT
Flame instability at <50% load	HHO fires first, anchors flame root at burner outlet	Stable operation confirmed at 20% rated load
High CO emissions at part load	OH radicals accelerate CO→CO ₂	CO decreased at all test loads with HHO
SO ₂ from sulphur-bearing coal	HHO inhibits S oxidation pathways	SO ₂ reduced from 142 ppm to 0 ppm (lean coal-SC blend)
Fly ash slagging (coarse particles)	More complete combustion, finer ash	PM>100µm reduced 10.32%
Lignite (low-calorific, wet coal)	H ₂ high flame speed ignites volatile-poor coal	Lignite furnace temperature increased; stability enhanced
Cold start / ignition	HHO at 0.02 mJ ignition energy vs 0.2 mJ hydrocarbon	10× easier ignition; reduces auxiliary oil or gas start fuel

4.4 Biomass Boilers (Wood chips, Pellets, Agricultural residues, MSW)

Overview

Biomass boilers are the most variable in terms of fuel composition. Moisture content, volatile matter, ash content, and particle size all fluctuate. This creates real combustion instability challenges — with CO and smoke frequently exceeding limits. HHO produces the most dramatic measured results in biomass applications.

★ Strongest Direct Evidence — HHO in Biomass Boiler (ScienceDirect, National Science Foundation of China)

Study: 'The pollutant discharge improvement by introducing HHO gas into biomass hot air generator', ScienceDirect, 2021. Supported by National Science Foundation of China (Grant No. 51806021).

- Platform: Medium-sized HHO generator introduced into biomass hot air generator (boiler) for mixed combustion
- HHO generator specifications: high-efficiency gas-liquid convection channel, 1.5mm plate spacing, air-cooling — adjustable gas output

Results measured:

- **CO reduction: 93.0%**
- **NO reduction: 22.5%**
- **Smoke reduction: 80%**
- Total operating cost during preheating: REDUCED based on electricity + biomass fuel economics
- 'Integration of biomass fuel and HHO gas can effectively reduce pollutant emissions and save fuel, especially in areas rich in renewable energy'

📌 Source: ScienceDirect, 2021 — 'The pollutant discharge improvement by introducing HHO gas into biomass hot air generator.' National Science Foundation of China Grant 51806021. Hunan Hydrogen Times Co. Ltd. provided HHO generators.

Why biomass benefits so dramatically: Biomass combustion inherently produces high CO because moisture in the fuel cools the flame below the CO oxidation threshold (~600°C). HHO raises the local flame temperature AND provides OH radicals that drive CO oxidation even at lower temperatures. The 93% CO reduction confirms this mechanism operating at its most effective.

BIOMASS FUEL TYPE	COMBUSTION CHALLENGE	HHO BENEFIT	SOURCE
Wood chips (high moisture 30–50%)	Moisture cools flame; high CO; poor ignition	OH radicals oxidise CO even at reduced temps; H ₂ improves ignition	NSFOC Study 2021
Wood pellets (dry,	Generally good; CO/NOx	Further CO/NOx reduction for	Literature consensus

consistent)	compliance target	compliance headroom	
Agricultural residues (rice husk, bagasse)	High ash; variable calorific value; flame instability	HHO stabilises flame; reduces ash carryover from better burnout	Extrapolated from coal/biomass studies
MSW / RDF (Municipal Solid Waste)	Highly variable composition; toxic emissions risk	More complete combustion; CO reduction reduces risk of chlorinated compound formation	Engineering inference from mechanism
Biogas (anaerobic digestion)	High CO ₂ dilution; reduced flame speed	H ₂ addition restores flame speed; stabilises lean combustion	DOE Lean H ₂ Combustion data

SECTION 5 — CLEANER BOILERS: THE SOOT REDUCTION BENEFIT IN DETAIL

One of the most commercially significant — yet least-publicised — benefits of HHO in boilers is the reduction of soot formation and deposition on heat transfer tube surfaces. This directly reduces maintenance burden, extends cleaning intervals, and sustains higher efficiency between service cycles.

5.1 The Soot Formation Mechanism & How HHO Interrupts It

Soot forms when: Fuel is not fully oxidised in the combustion zone. Carbon that should become CO₂ instead forms carbonaceous particles (C_{2-n}, PAH, and ultimately soot). The primary cause is insufficient OH radical concentration in the flame to complete the oxidation chain.

Without HHO:

- Conventional kerosene/coal/biomass flame has limited OH radical generation in the post-flame zone
- Unburned carbon particles survive the flame and travel with flue gas
- These particles deposit on cooler tube surfaces in the convective section (superheater, economiser, air preheater)
- 1mm soot layer = 5–10% heat transfer loss and 8–10% more fuel needed

With HHO:

- HHO combustion generates abundant OH radicals (via $H + O_2 \rightarrow OH + O$)
- OH attacks and oxidises carbon intermediates before they can form soot particles
- Less unburned carbon = fewer particles entering the flue gas stream
- Less soot deposition on tube surfaces = sustained heat transfer efficiency
- Less smoke in stack = lower particulate emissions (80% smoke reduction confirmed in biomass test)

5.2 Quantified Impact of Soot on Boiler Economics — Then HHO's Role

SCENARIO	SOOT THICKNES S	EFFICIENCY LOSS	ANNUAL FUEL WASTE (₹ basis 10 T/H boiler at ₹60/kg coal)	HHO PREVENTS THIS
Clean tubes (Day 1)	0 mm	0%	₹0 waste	—
2 weeks of operation without HHO	~0.8–1.0 mm	5–10%	₹18–36 lakhs/year	HHO reduces soot formation rate
1 month operation without cleaning	~2–3 mm	15–25%	₹54–90 lakhs/year	HHO extends time to this level

Pre-shutdown (2–3 months)	3–5 mm	25–40%	₹90–144 lakhs/year	HHO reduces cleaning frequency
With HHO: slower soot accumulation	Lower effective thickness	Efficiency sustained 3–7% higher	₹10–25 lakhs/year saved	Sustained clean combustion

Note: Calculations illustrative for a 10 T/H coal-fired steam boiler operating 8,000 hrs/year at 60 kg coal/T steam, ₹60/kg coal. Actual values depend on boiler size, fuel, and operating hours.

📌 Source: CED Engineering M06-022 — soot thickness vs efficiency loss data. Goodway Technologies — soot accumulation rate data. CoalBiomassBoiler.com — 1mm soot = 8–10% fuel increase. ScienceDirect HHO biomass 2021 — 80% smoke reduction = 80% less soot-forming particles.

5.3 Real Case: Refinery Boiler Fouling Without Intervention

Case Study Reference — 30 T/H Refinery Steam Boiler (CoalBiomassBoiler.com Engineering Data)

- Boiler capacity: 30 T/H steam generation
- Observation: Flue gas temperature rose from 190°C to 240°C over six months
- Root cause: Progressive soot and fouling accumulation on tube surfaces
- Impact: +50°C stack temperature = ~5–6% efficiency loss = significant fuel waste
- Industry rule: Every 40°F (22°C) rise in stack temperature = ~1% efficiency penalty
- HHO application: Sustained complete combustion would slow this stack temperature rise significantly by reducing soot precursor formation

📌 Source: CoalBiomassBoiler.com Engineering Reference — 'Top 5 Key Factors Affecting Industrial Gas Boiler Efficiency': 'In a refinery boiler system rated at 30 T/H steam capacity, flue gas temperature rose from 190°C to 240°C over six months.'

5.4 Laboratory Evidence on Hydrogen and Soot Reduction

ORNL Research — Soot Chemistry and Hydrogen's Role (Oak Ridge National Laboratory)

Oak Ridge National Laboratory (ORNL) research on furnace combustion and soot:

- 'Soot particles, which typically form because of the incomplete combustion of hydrocarbons, contain substantial hydrogen'
- ORNL research confirmed: complete combustion of hydrocarbons prevents soot particle formation — the mechanism that HHO facilitates
- ORNL's neutron imaging mapped soot accumulation patterns in heat exchangers — confirming that combustion completeness is the primary soot-formation control variable

📌 Source: Oak Ridge National Laboratory / Neutron Science at ORNL — 'Add-on device makes home furnaces cleaner, safer and longer-lasting': 'Soot particles, which typically form because of the incomplete combustion of hydrocarbons, contain substantial hydrogen.'

SECTION 6 — APPLICABILITY ACROSS BOILER SIZES & CONFIGURATIONS

SGT's HHO generator range scales from small industrial units (200 LPH) to large multi-generator configurations (3,000+ LPH). The technology applies across all standard boiler configurations.

BOILER CATEGORY	CAPACITY RANGE	FUEL TYPES	HHO SIZING	PRIMARY BENEFITS	REFERENCE
Small fire-tube / shell boiler	0.5–5 T/H steam 50–500 kW	NG, LPG, LDO, diesel	200–500 LPH	5–10% fuel saving; CO/NOx reduction; cleaner tubes	LPG boiler IJHE 2024; industrial burner Fuel 2025
Medium package water-tube boiler	5–20 T/H steam 0.5–2 MW	NG, LPG, HFO, dual-fuel	500–1500 LPH	Combustion completeness; reduced cleaning; emissions compliance	MDPI Energies 2025; Fuel 2025
Large industrial boiler	20–100 T/H steam 2–10 MW	Coal, HFO, biomass, NG	1500–5000 LPH	Flame stability; coal low-load operation; SO ₂ reduction	ScienceDirect 2024 coal studies
Power plant boiler (utility scale)	100–1000+ MW	Pulverised coal, coal blends	Multi-generator arrays	Ultra-low-load stabilisation; NOx/CO compliance; peak-shaving support	ScienceDirect Applied Energy 2024
Biomass/ waste-to-energy boiler	Variable 1–50 T/H	Biomass, MSW, RDF, biogas	Matched to fuel variability	93% CO reduction; 80% smoke reduction; stable flame under variable fuel	ScienceDirect NSFOC 2021
Condensing boiler (residential/commercial)	10–500 kW	NG, LPG	100–300 LPH	H ₂ blend compatibility; CO 88% reduction; NOx 75% reduction	MDPI Energies 2025; ScienceDirect 2024

6.1 Babcock & Wilcox (B&W) — Industrial Precedent for Hydrogen in Boilers

Babcock & Wilcox (B&W) — one of the world's leading boiler manufacturers — has operated hydrogen-fired industrial boilers commercially:

- 'B&W has supplied more than 5,000 water-tube package and industrial boilers with an established reputation for reliability'
- 'We have used our experience with a wide range of fuels to supply more than 60 hydrogen steam boilers and units firing hydrogen-blended fuels'
- 'With the anticipated demand for wider utilisation of cleaner-burning hydrogen and other low- or no-carbon fuels, B&W is ready to advance steam generation'
- This establishes that hydrogen-in-boiler technology is proven at industrial scale by a major OEM — validating the scientific principles that SGT's HHO catalyst leverages

📌 Source: Babcock & Wilcox (B&W) BrightGen™ Hydrogen Steam Boiler programme — babcock.com/home/environmental/decarbonization/hydrogen-combustion

SECTION 7 — CONSOLIDATED PERFORMANCE DATA & REFERENCED EVIDENCE

7.1 Performance Summary Across All Fuel Types

FUEL TYPE	BOILER TYPE	CO REDUCTION	NOx REDUCTION	FUEL SAVING	SMOKE/ PARTICULATE	SOURCE
Natural Gas (CNG)	Condensing boiler	88% (at 20% H ₂ blend)	75%	~3–7%	Near zero unburned HC	MDPI Energies 2025
Natural Gas	Industrial burner 40,000 kcal/h	25%	—	~5% (80.6% efficiency)	7% HC reduction	Elsevier Fuel 2025
LPG	Burner/boiler system	Improved (measured)	Improved	Improved (measured)	Improved	IJHE 2024
Light Oil / Kerosene	Oil-fired boiler	Improved	Reduced	~5–8%	Reduced	MDPI Energies 10:251
Coal (bituminous)	200kW pulverised furnace	Decreased	Optimised by staging	Efficiency improved	Finer fly ash	Applied Energy 2024
Coal (lean coal)	200kW furnace	Decreased	SO ₂ : 0 ppm (from 142)	Stability enables full load	Finer particles	Applied Energy 2024
Coal (lignite)	200kW furnace	Decreased	—	Stability improved	—	Applied Energy 2024
Biomass (mixed)	Hot air generator/boiler	93% CO reduction	22.5%	Cost reduced	80% smoke reduction	ScienceDirect NSFOC 2021

7.2 Complete Research Reference List

#	REFERENCE	KEY FINDING	JOURNAL / DATE
1	MDPI Energies 2025 — H ₂ -enriched NG condensing boiler	20% H ₂ : CO -88%, NO _x -75%; CO+OH→CO ₂ mechanism confirmed	MDPI Energies 18(13):3240, June 2025
2	Elsevier Fuel Journal 2025 — HHO industrial burner	HHO: +8-9°C flame temp, -25% CO, -7% HC, efficiency 80.6%	Elsevier Fuel Journal, Jan 2025
3	MDPI Energies 10:251 — HHO + coal + oil in boiler	HHO with coal and oil improves boiler electricity generation efficiency	MDPI Energies, Vol 10, Issue 2, 2017
4	ScienceDirect 2021 — HHO biomass boiler (NSFOC)	CO -93%, NO -22.5%, smoke -80% in biomass hot air generator	ScienceDirect, 2021; NSFOC Grant 51806021

5	ScienceDirect Applied Energy 2024 — HHO coal pulverised furnace	HHO assists ultra-low-load combustion of 3 coal types; SO ₂ 142→0 ppm	Applied Energy / ScienceDirect, 2024
6	J. Energy Institute 2024 — HHO coal co-firing modes	Air staging + HHO optimises low-load stability and emissions	Journal of Energy Institute, 2024
7	ScienceDirect Case Studies 2025 — HHO low-load NOx	HHO improves NOx and combustion stability at 20-50% boiler load	Case Studies in Thermal Engineering, 2025
8	ScienceDirect 2024 — Porous burner H ₂ boiler	CO 154→18 ppm; NO 20→5 ppm with H ₂ enrichment	ScienceDirect, 2024
9	ResearchGate / IJHE 2024 — LPG boiler efficiency	HHO addition improves LPG boiler efficiency; water heating rate improved	Int. Journal of Hydrogen Energy, 2024
10	OSTI/DOE — Lean Hydrogen Combustion Ch. 8	10% H ₂ extends lean blowout 10.4%; 20% H ₂ by 16.6%	US DOE OSTI Publication 1731098, 2021
11	Springer — H ₂ /NG/air laminar flame speed	H ₂ max flame speed 2.933 m/s vs NG 0.374 m/s	Frontiers Chem. Eng., Huang et al., 2010
12	ASME J. Gas Turbines Power 2021 — Kerosene flame speed	Kerosene laminar flame speed ~0.376 m/s	ASME JGTP, 2021
13	ScienceDirect Chem. Eng. Sci. 2023 — n-decane/H ₂	Key reaction H+O ₂ =OH+O; LBV and AFT enhanced; NO reduced 24%	ScienceDirect CES, 2023
14	ScienceDirect Fuel 2025 — H ₂ oxy-fuel burner scale-up	H ₂ oxy-fuel flame: 2800K max temp, 0.15m length	ScienceDirect Fuel, 2025
15	CED Engineering M06-022 — Boiler efficiency	0.8mm soot = 9.5% HT loss; 4.5mm = 69%; 40°F rise = 1% efficiency loss	CED Engineering, M06-022
16	Goodway Technologies — Soot data	3.2mm soot = 47% heat loss + 8.5% fuel increase	Goodway Technologies Engineering Data
17	Nicholls & Augustine, Bureau of Mines, 1935	1/16 inch soot = 4.2% efficiency loss; 1/4 inch = 7%	Bureau of Mines Technical Paper, 1935
18	CoalBiomassBoiler.com — Gas/Coal/Biomass efficiency	1mm soot = 8–10% fuel increase; 30T/H refinery case	Industry engineering references, 2024–2025
19	Babcock & Wilcox BrightGen™	>60 H ₂ steam boilers and H ₂ -blend units supplied commercially	B&W babcock.com, 2024
20	ORNL Neutron Science — Soot and furnace cleaning	Soot from incomplete combustion; H prevents soot; ORNL neutron mapping	Oak Ridge National Laboratory, 2024

SECTION 8 — FINANCIAL IMPACT MODEL FOR INDUSTRIAL BOILER OPERATORS

8.1 The Two-Stream ROI: Fuel Savings + Maintenance Savings

HHO's ROI for boiler operators comes from two compounding streams — direct fuel savings and indirect maintenance savings from reduced soot accumulation. Most customers focus on fuel; maintenance savings are often comparable in magnitude.

BOILER SCENARIO	ANNUAL FUEL COST (ESTIMATE)	FUEL SAVING (5%)	MAINTENANCE SAVING (ESTIMATE)	COMBINED ANNUAL BENEFIT
5 T/H gas-fired steam boiler, NG at ₹50/SCM, 6000 hrs/year	₹45,00,000	₹2,25,000	₹80,000–1,50,000 (fewer cleaning cycles)	₹3,05,000–3,75,000/year
10 T/H oil-fired boiler, HFO at ₹65/litre, 7000 hrs/year	₹1,82,00,000	₹9,10,000	₹2,00,000–3,50,000 (soot + tube cleaning)	₹11,10,000–12,60,000/year
20 T/H coal-fired boiler, coal at ₹8/kg, 7500 hrs/year	₹1,20,00,000	₹6,00,000	₹1,50,000–3,00,000 (soot blowing + downtime)	₹7,50,000–9,00,000/year
50 T/H biomass boiler, biomass at ₹5/kg, 7500 hrs/year	₹93,75,000	₹4,69,000	₹2,00,000–4,00,000 (CO compliance, tube cleaning)	₹6,69,000–8,69,000/year

Note: Estimates based on 5% fuel saving (conservative mid-range). Maintenance savings based on reduced cleaning frequency (HHO reduces soot formation rate, extending tube cleaning intervals). Actual savings depend on operating hours, fuel quality, and boiler condition.

8.2 SGT HHO Generator System — Operational Cost

SGT HHO generators are engineered for 24×7 continuous industrial operation:

- Electricity consumption: ~400W per 130 LPH of HHO output (ScienceDirect 2025: 2 Wh/L energy intensity)
- Water consumption: ~0.9 litres of water per 1,000 litres of HHO produced (stoichiometric)
- Typical industrial HHO dosing: 100–500 LPH for boilers 5–50 T/H capacity
- Operating cost at 400W, ₹8/kWh: ~₹3.2/hour — vs fuel savings of ₹50–300/hour
- Return ratio: typically 15:1 to 60:1 (fuel saved per ₹ electricity spent on HHO)
- Payback period: typically 8–18 months depending on boiler size, fuel type, and operating hours

📌 Source: ScienceDirect Fuel Journal 2025 — 'Optimum operation at 10–15A (400W) produced 130 L/h HHO with 2 Wh/L energy intensity.'

SECTION 9 — SGT HYDROEDGE SOLUTION: IMPLEMENTATION FOR BOILERS

9.1 System Overview

SGT HydroEdge industrial HHO generators are patent-applied, Made in India systems designed for 24×7 continuous industrial operation. Unlike vehicle-grade or Chinese mass-produced units, SGT systems are built for the thermal loads, duty cycles, and safety standards of industrial boiler rooms.

FEATURE	SGT SPECIFICATION	ENGINEERING RATIONALE
HHO output range	200 LPH to 3,000 LPH (single unit)	Covers boilers from 1 T/H to 50+ T/H
Duty cycle	24×7 continuous operation certified	Industrial boilers run continuously; generator must match
Fuel compatibility	Gas, LPG, LDO, HFO, diesel, coal, biomass — all	Single platform across all boiler fuel types
Control system	IoT-enabled real-time dosing control	GreenVision Digital Twin: adjusts HHO flow with boiler load
Safety systems	Flashback arrestor, pressure relief, dry-cell design, auto water management	Industrial boiler room safety compliance
Scalability	Multiple units stackable for large boilers	No upper limit — scales to utility boiler applications
Monitoring & reporting	GreenVision platform: fuel savings, emissions, uptime	Investor-grade data for ESG reporting and ROI verification
Made in India	Patent-applied technology; only non-Chinese manufacturer of industrial HHO	Supply chain security; domestic service support; Atmanirbhar Bharat

9.2 Implementation Protocol

SGT's 4-Stage Boiler Implementation

- **STAGE 1 — BASELINE AUDIT (Week 1–2):** Measure current combustion efficiency (flue gas O₂, CO, CO₂, stack temperature), fuel consumption, and tube cleanliness status. Establish the performance baseline against which savings will be measured.
- **STAGE 2 — INSTALLATION (Week 2–3):** HHO generator installed at burner air/fuel inlet. IoT sensors connected to GreenVision platform. No modification to existing boiler — purely additive installation.
- **STAGE 3 — OPTIMISATION (Week 4–8):** HHO flow rate optimised against boiler load profile. GreenVision Digital Twin calibrates optimal dosing for peak efficiency. First combustion analysis report issued to customer.
- **STAGE 4 — CONTINUOUS MONITORING:** Monthly performance reports. Annual tube

inspection protocol to verify soot reduction. IoT data available to customer for ESG reporting and investor disclosure.

SGT HydroEdge — Industrial HHO Generators | Made in India | Patent Applied

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GreenLabs — Sustainability Infrastructure for Industry | GreenVision Digital Twin Platform